

**Project Design Phase-II
Technology Stack (Architecture & Stack)**

Date	26 June 2026
Team ID	
Project Name	Metro Ticket Generating System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table 1: Components & Technologies

Metro Ticket Generating System – Architecture Components

S.No	Component	Description	Technology
1	User Interface	Passenger Portal and Administrator Portal for ticket booking, station management, and booking history.	ServiceNow Service Portal / Workspace / Forms / Lists
2	Application Logic	Handles user authentication, ticket booking, fare calculation, payment validation, QR ticket generation, and booking status updates.	Business Rules, Client Scripts, Script Includes, UI Policies
3	Workflow Automation	Automates ticket booking workflow, payment confirmation, QR ticket generation, and notification process.	Flow Designer
4	Database	Stores passenger information, station details, metro routes, fare rules, ticket bookings, payment records, and QR ticket data.	sys_user, Metro Ticket Table, Station Table, Route Table, Fare Table (Custom Tables)
5	Email & Notification Service	Sends booking confirmation, payment confirmation, QR ticket, cancellation, and reminder notifications to passengers.	ServiceNow Notifications, Email Service

Metro Ticket Generating System

Mini Architecture Diagram — ServiceNow Platform

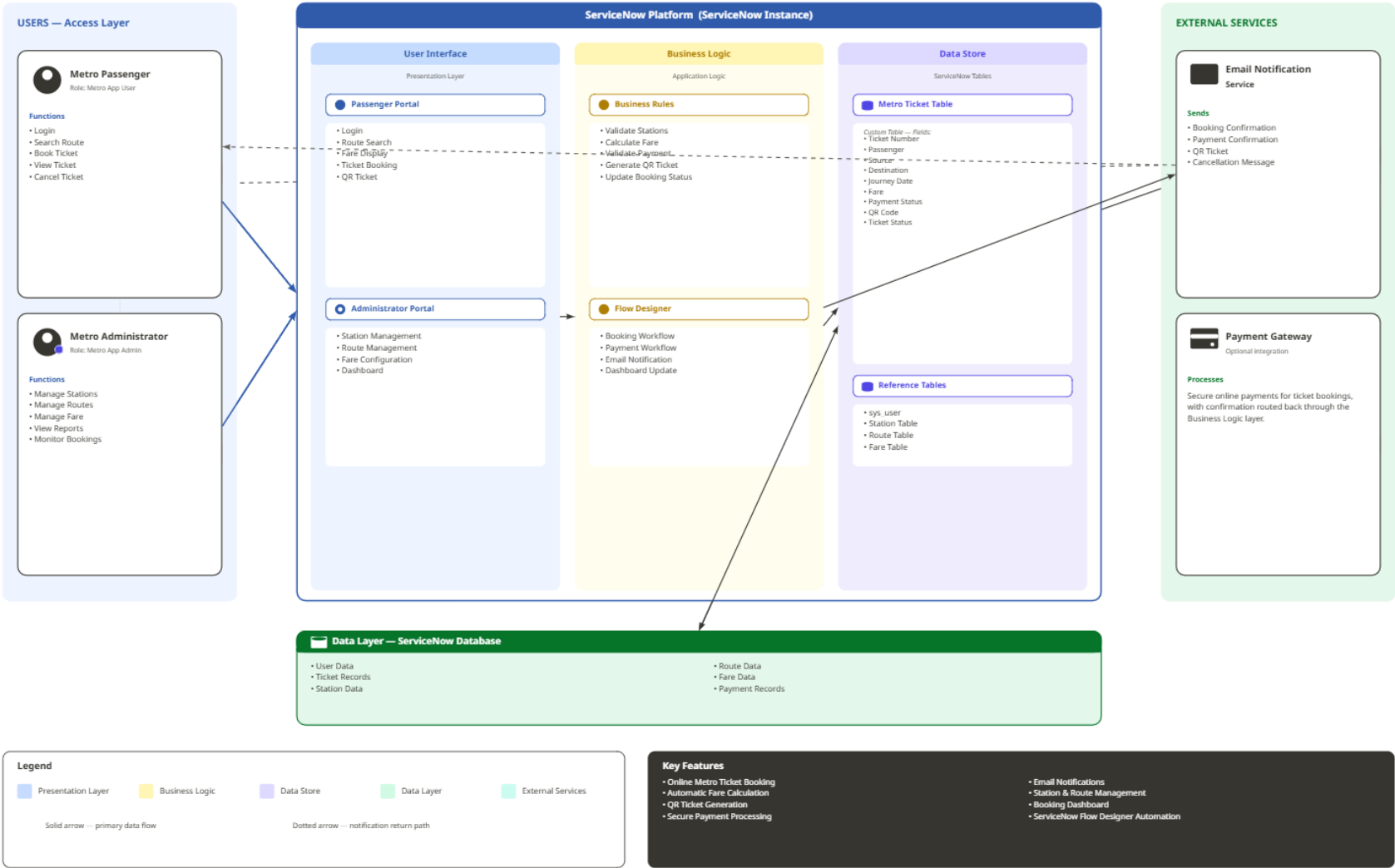


Table 2: Application Characteristics

System Characteristics – Metro Ticket Generating System

Characteristic	Description	Technology
Security	Role-based access ensures that passengers can access only their own ticket bookings, while administrators can manage stations, routes, fares, and bookings.	ACLs, Roles (x_metro_app.user, x_metro_app.admin)
Scalability	Supports increasing numbers of passengers, metro stations, routes, and ticket bookings without major architectural changes.	ServiceNow Platform
Availability	The application is available whenever the ServiceNow instance is operational, enabling passengers to book tickets anytime.	ServiceNow Cloud
Performance	Provides fast ticket booking, fare calculation, QR ticket generation, and workflow execution with minimal response time.	Flow Designer, Business Rules
Reliability	Maintains accurate ticket booking records, payment status, QR ticket data, and notifications without data loss.	ServiceNow Database
Usability	Offers a simple, intuitive interface for passengers and administrators to perform booking and management tasks efficiently.	Service Portal, Workspace, Forms & Lists
Maintainability	Allows easy updates to stations, routes, fare rules, and workflows through configurable ServiceNow components.	Flow Designer, Script Includes, Business Rules
Interoperability	Supports integration with external payment gateways and notification services for seamless ticket booking.	REST APIs, IntegrationHub